

*“In theory, there is no difference
between
theory and practice. In practice,
there is.”*

— Yogi Berra (attributed)

Abstract

Multiagent AI systems—autonomous coding assistants, research pipelines, financial decision engines—have moved from prototype to production in under two years. With them comes a new class of security concern: attacks that target not data or infrastructure but the *reasoning processes* of AI agents. Prompt injections that propagate through delegation chains, trust relationships that launder adversarial influence, and coordination mechanisms vulnerable to strategic manipulation all represent cognitive attack surfaces absent from traditional security models.

The **Cognitive Integrity Framework (CIF)**, presented across two companion papers, offers the first comprehensive formal and computational treatment of this problem. Part 1 establishes mathematical foundations: a trust calculus with provably bounded delegation, defense composition algebras with multiplicative detection guarantees, and information-theoretic limits on attack stealth. Part 2 provides computational validation: eight implemented defense modules (1,594 passing tests), a 950-attack corpus spanning four threat categories, and parametric